FYBCA Semester 2 — Complete Exam Notes

Subject 3: Basic Mathematics for Computing

KBCNMU | Exam: 30 Marks | Complete Notes

TOPIC 1: Mathematical Logic — Statements and Truth Values

1.1 What is Mathematical Logic?

Mathematical Logic is a branch of mathematics that studies the structure of reasoning using formal symbols. It allows us to evaluate whether arguments are valid or invalid.

1.2 Statement (Proposition)

A STATEMENT (also called PROPOSITION) is a declarative sentence that is either TRUE or FALSE — but NOT BOTH at the same time.

Example	Is it a Statement?	Truth Value
The sun rises in the east.	Yes	True (T)
2 + 2 = 5	Yes	False (F)
What time is it?	No — it is a question	Not a statement
x + 1 = 5	No — truth depends on x	Not a proposition (open statement)

1.3 Types of Statements

Type	Definition	Example
Simple Statement	Cannot be broken into simpler parts.	'It is raining.'
Compound Statement	Formed by combining two or more simple statements using connectives.	'It is raining AND it is cold.'
Negation	The opposite of a statement. If P is true, ~P is false.	P: 'It is hot.' ~P: 'It is NOT hot.'

1.4 Logical Connectives

Connective	Symbol	Name	Meaning
NOT	~ or ¬	Negation	Reverses truth value
AND	∧	Conjunction	Both must be true
OR	∨	Disjunction	At least one must be true
IF...THEN	→	Conditional / Implication	If P then Q
IF AND ONLY IF	↔	Biconditional	Both same truth value

TOPIC 2: Truth Tables

2.1 Negation (~P)

P	~P
T	F
F	T

2.2 Conjunction (P ∧ Q) — AND

P	Q	P ∧ Q
T	T	T
T	F	F
F	T	F
F	F	F

AND is TRUE only when BOTH P and Q are TRUE.

2.3 Disjunction (P ∨ Q) — OR

P	Q	P ∨ Q
T	T	T
T	F	T
F	T	T
F	F	F

OR is FALSE only when BOTH P and Q are FALSE.

2.4 Conditional (P → Q) — IF THEN

P	Q	P → Q
T	T	T
T	F	F
F	T	T
F	F	T

Conditional is FALSE only when P is TRUE and Q is FALSE. (Promise broken only when condition met but conclusion not followed.)

2.5 Biconditional (P ↔ Q)

P	Q	P ↔ Q
T	T	T
T	F	F
F	T	F

F	F	T

Biconditional is TRUE only when both P and Q have the SAME truth value.

2.6 Complete Truth Table Example

Example: Construct truth table for (P ∧ Q) → (~P ∨ R)

P	Q	R	P∧Q	~P	~P∨R	(P∧Q)→(~P∨R)
T	T	T	T	F	T	T
T	T	F	T	F	F	F
T	F	T	F	F	T	T
T	F	F	F	F	F	T
F	T	T	F	T	T	T
F	T	F	F	T	T	T
F	F	T	F	T	T	T
F	F	F	F	T	T	T

Memory Tricks

• AND (∧): BOTH must be true — like a strict gatekeeper.
• OR (∨): AT LEAST ONE must be true — like a lenient gatekeeper.
• NOT (~): Flips the truth value.
• IF-THEN (→): Only FALSE when P=T and Q=F (broken promise).
• BICONDITIONAL (↔): TRUE only when both same (TT or FF).

TOPIC 3: Tautology and Contradiction

3.1 Tautology

A TAUTOLOGY is a compound statement that is ALWAYS TRUE regardless of the truth values of its component statements.

Example: $P \vee \sim P$ is always TRUE (Law of Excluded Middle):

P	~P	P ∨ ~P
T	F	T
F	T	T

Other tautologies: $(P \to Q) \leftrightarrow (\sim Q \to \sim P)$, $P \to (Q \to P)$

3.2 Contradiction (Fallacy)

A CONTRADICTION is a compound statement that is ALWAYS FALSE regardless of the truth values of its component statements.

Example: $P \wedge \sim P$ is always FALSE:

P	~P	P ∧ ~P
T	F	F
F	T	F

3.3 Contingency

A CONTINGENCY is a statement that is neither a tautology nor a contradiction — it is sometimes TRUE and sometimes FALSE depending on values.

Type	Definition	Result
Tautology	Always TRUE for all truth value combinations	All T in final column
Contradiction	Always FALSE for all truth value combinations	All F in final column
Contingency	Sometimes TRUE, sometimes FALSE	Mix of T and F in final column

Must Remember

- Tautology: ALWAYS TRUE — all T in last column of truth table.
- Contradiction: ALWAYS FALSE — all F in last column.
- Contingency: Mix of T and F — neither tautology nor contradiction.
- Logical Equivalence: Two statements with same truth table are logically equivalent.

Exam Questions

- Define tautology and contradiction. Give examples. (4 marks)
- Construct truth table for $P \to Q$ and check if tautology. (4 marks)
- What is a compound statement? Explain logical connectives. (6 marks)

TOPIC 4: Sets and Types of Sets

4.1 What is a Set?

A SET is a well-defined collection of distinct objects. The objects in a set are called ELEMENTS or MEMBERS. Sets are denoted by capital letters (A, B, C) and elements are written within curly braces {}.

Example: A = {1, 2, 3, 4, 5} — Set A containing integers 1 to 5.

4.2 Set Notation

Symbol	Meaning	Example
∈	Element belongs to set	3 ∈ A
∉	Element does not belong	6 ∉ A
|A| or n(A)	Cardinality (number of elements)	If A={1,2,3}, |A|=3
{ }	Empty set (no elements)	{ } or ∅

4.3 Types of Sets

Type	Definition	Example
Empty Set (Null Set)	Set with NO elements. Denoted ∅ or {}	∅ = {} = set of people over 200 years old
Finite Set	Set with countable number of elements	A = {1, 2, 3, 4, 5}
Infinite Set	Set with uncountable elements	Set of all natural numbers
Singleton Set	Set with exactly ONE element	B = {5}
Universal Set	Set containing ALL elements under consideration. Denoted U	U = {1,2,3,...,100}
Subset	If every element of A is in B, then A⊆B	A={1,2}, B={1,2,3}: A⊆B
Proper Subset	A⊂B: A⊆B but A≠B	A={1,2}⊂B={1,2,3}
Power Set	Set of all subsets of A. Denoted P(A). If |A|=n, |P(A)|=2^n	A={1,2}: P(A)={∅,{1},{2},{1,2}}
Equal Sets	Two sets with exactly same elements	A={1,2}=B={2,1}
Equivalent Sets	Sets with same cardinality (same number of elements)	A={1,2,3} and B={a,b,c} are equivalent
Disjoint Sets	Sets with NO common elements	A={1,2} and B={3,4} are disjoint

TOPIC 5: Set Operations and Venn Diagrams

5.1 Set Operations

Operation	Symbol	Definition	Example
Union	$A \cup B$	Set of all elements in A or B or both	A={1,2,3}, B={3,4,5}: A∪B={1,2,3,4,5}
Intersection	$A \cap B$	Set of elements common to both A and B	A={1,2,3}, B={3,4,5}: A∩B={3}
Difference	$A - B$	Elements in A but NOT in B	A={1,2,3}, B={3,4}: A-B={1,2}
Complement	A' or A^c	Elements in Universal Set but NOT in A	U={1..5},A={1,2}: A'={3,4,5}
Symmetric Diff.	$A \oplus B$	Elements in A or B but NOT in both	A={1,2,3},B={3,4}: A⊕B={1,2,4}

5.2 Laws of Set Theory

Law	Formula
Commutative Law	$A \cup B = B \cup A$ and $A \cap B = B \cap A$
Associative Law	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributive Law	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity Law	$A \cup \emptyset = A$ and $A \cap U = A$
Complement Law	$A \cup A' = U$ and $A \cap A' = \emptyset$
De Morgan's Law	$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$
Idempotent Law	$A \cup A = A$ and $A \cap A = A$
Double Complement	$(A')' = A$

5.3 Cartesian Product

The CARTESIAN PRODUCT of sets A and B, written $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

If $|A| = m$ and $|B| = n$, then $|A \times B| = m \times n$

Example: A = {1, 2} and B = {a, b, c}

A × B = {(1,a), (1,b), (1,c), (2,a), (2,b), (2,c)} — 2 × 3 = 6 pairs

Note: $A \times B \neq B \times A$ (ordered pairs — order matters!)

5.4 Venn Diagrams

Venn diagrams use overlapping circles to represent sets and their relationships. The universal set U is represented by a rectangle, each set by a circle inside it.

- $A \cup B$: ENTIRE shaded area of both circles.
- $A \cap B$: Only the OVERLAPPING region.
- $A - B$: Left circle MINUS the overlap.

- A': Everything OUTSIDE circle A (inside rectangle).

Important Formulas

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ [Inclusion-Exclusion Principle]
- $n(A \times B) = n(A) \times n(B)$
- P(A) has 2^n subsets where n = |A|
- De Morgan: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$
- For disjoint sets: $n(A \cap B) = 0$, so $n(A \cup B) = n(A) + n(B)$

Exam Questions

- Define set. What are different types of sets? Explain with examples. (6 marks)
- Explain set operations with examples. (6 marks)
- State and explain De Morgan's Laws. (4 marks)
- What is Cartesian Product? Give example. (4 marks)
- If $n(A)=20$, $n(B)=15$, $n(A \cup B)=30$, find $n(A \cap B)$. (2 marks)

TOPIC 6: Determinants — Properties and Evaluation

6.1 What is a Determinant?

A DETERMINANT is a scalar (single number) value associated with every SQUARE matrix. It is denoted as det(A) or |A|.

6.2 Determinant of 2×2 Matrix

For matrix A = [[a, b], [c, d]]:

|A| = ad - bc

Example: A = [[3, 2], [1, 4]] => |A| = (3)(4) - (2)(1) = 12 - 2 = 10

6.3 Determinant of 3×3 Matrix (Cofactor Expansion)

For 3×3 matrix A = [[a1,b1,c1],[a2,b2,c2],[a3,b3,c3]], expand along first row:

|A| = a1(b2c3 - b3c2) - b1(a2c3 - a3c2) + c1(a2b3 - a3b2)

6.4 Minors and Cofactors

MINOR M_ij: Determinant of the matrix obtained by deleting row i and column j.

COFACTOR A_ij: A_ij = (-1)^(i+j) × M_ij

Sign pattern for cofactors:

	Col 1	**Col 2**	**Col 3**
Row 1	+	-	+
Row 2	-	+	-
Row 3	+	-	+

6.5 Properties of Determinants

Property	**Statement**
1. Zero Row	If all elements of a row/column are 0, determinant = 0
2. Identical Rows	If two rows/columns are identical, determinant = 0
3. Row Swap	If two rows/columns are interchanged, sign of determinant changes
4. Scalar Multiple	If one row/column is multiplied by k, determinant is multiplied by k
5. Row Addition	Adding multiple of one row to another does NOT change determinant
6. Transpose	det(A) = det(A^T) — transpose doesn't change determinant
7. Product	det(AB) = det(A) × det(B)

6.6 Worked Example — 3×3 Determinant

Find determinant of A = [[1,2,3],[4,5,6],[7,8,9]] Expanding along Row 1:

$|A| = 1\times|5,6;8,9| - 2\times|4,6;7,9| + 3\times|4,5;7,8|$

$= 1\times(45-48) - 2\times(36-42) + 3\times(32-35)$

$= 1\times(-3) - 2\times(-6) + 3\times(-3)$

$= -3 + 12 - 9 = 0$

Important Points

- Determinant is defined ONLY for square matrices (2x2, 3x3, etc.).
- If det(A) = 0, matrix is called SINGULAR (no inverse).
- If det(A) ≠ 0, matrix is called NON-SINGULAR (has inverse).
- Minor = delete row i and column j, take determinant.
- Cofactor = Minor with appropriate sign: $(-1)^{(i+j)} \times$ Minor.

TOPIC 7: Cramer's Rule

7.1 What is Cramer's Rule?

Cramer's Rule is a method to solve a system of LINEAR EQUATIONS using DETERMINANTS. It is applicable when the number of equations equals the number of unknowns AND the coefficient determinant is NOT zero.

7.2 Cramer's Rule for 2 Variables

System: $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$

Step 1: Find coefficient determinant $D = |a_1,b_1;a_2,b_2| = a_1b_2 - a_2b_1$

Step 2: $D_x = |c_1,b_1;c_2,b_2| = c_1b_2 - c_2b_1$

Step 3: $D_y = |a_1,c_1;a_2,c_2| = a_1c_2 - a_2c_1$

Step 4: $x = D_x/D$ and $y = D_y/D$ (valid only if $D \neq 0$)

7.3 Solved Example

Solve: $2x + 3y = 8$ and $4x - y = 2$

$D = (2)(-1) - (4)(3) = -2 - 12 = -14$

$D_x = (8)(-1) - (2)(3) = -8 - 6 = -14$

$D_y = (2)(2) - (4)(8) = 4 - 32 = -28$

$x = D_x/D = -14/-14 = 1$

$y = D_y/D = -28/-14 = 2$

Solution: $x = 1$, $y = 2$

7.4 Cramer's Rule for 3 Variables

System: $a_1x + b_1y + c_1z = d_1$ (and two more equations)

D = det(coefficient matrix)

D_x = replace column 1 (x-column) with constants d_1,d_2,d_3

D_y = replace column 2 (y-column) with constants d_1,d_2,d_3

D_z = replace column 3 (z-column) with constants d_1,d_2,d_3

Solution: $x = D_x/D$, $y = D_y/D$, $z = D_z/D$

Must Remember

- Cramer's Rule requires: same number of equations and unknowns.
- Condition: $D \neq 0$ (otherwise no unique solution).
- Dx: Replace first column of D with constant terms.
- Dy: Replace second column of D with constant terms.
- Dz: Replace third column of D with constant terms.

Exam Questions

- State Cramer's Rule. Use it to solve 2 linear equations with 2 unknowns. (6 marks)
- Solve using Cramer's Rule: $3x+2y=7$, $x-y=1$. (4 marks)
- Solve using Cramer's Rule system of 3 equations with 3 unknowns. (6 marks)

TOPIC 8: Matrices — Types and Operations

8.1 What is a Matrix?

A MATRIX is a rectangular array of numbers arranged in ROWS and COLUMNS. It is denoted by a capital letter. A matrix with m rows and n columns is called an m×n matrix.

Example: A 2×3 matrix: A = [[1,2,3],[4,5,6]] (2 rows, 3 columns)

8.2 Types of Matrices

Type	Description	Example
Row Matrix	Only ONE row	[1, 2, 3]
Column Matrix	Only ONE column	[[1],[2],[3]]
Square Matrix	Rows = Columns (m = n)	[[1,2],[3,4]] (2×2)
Zero/Null Matrix	All elements are 0	[[0,0],[0,0]]
Identity Matrix (I)	Square matrix with 1s on diagonal, 0s elsewhere	[[1,0],[0,1]]
Diagonal Matrix	Square matrix with non-zero elements only on main diagonal	[[3,0],[0,5]]
Scalar Matrix	Diagonal matrix with ALL diagonal elements equal	[[4,0],[0,4]]
Symmetric Matrix	A = A^T (same as its transpose)	[[1,2],[2,3]]
Skew-Symmetric	A^T = -A	[[0,3],[-3,0]]
Upper Triangular	All elements BELOW main diagonal are 0	[[1,2,3],[0,4,5],[0,0,6]]
Lower Triangular	All elements ABOVE main diagonal are 0	[[1,0,0],[2,3,0],[4,5,6]]
Singular Matrix	det(A) = 0 (no inverse exists)	det = 0
Non-Singular Matrix	det(A) ≠ 0 (inverse exists)	det ≠ 0

8.3 Matrix Operations

Addition and Subtraction

Matrices must have the SAME dimensions. Add/subtract corresponding elements.

If A = [[1,2],[3,4]] and B = [[5,6],[7,8]], then A+B = [[6,8],[10,12]]

Scalar Multiplication

Multiply each element of matrix by the scalar (constant).

If k=2 and A = [[1,2],[3,4]], then 2A = [[2,4],[6,8]]

Matrix Multiplication

A (m×n) × B (n×p) = C (m×p). The number of columns of A must equal number of rows of B.

C_ij = sum of products of i-th row of A and j-th column of B.

Example: A=[[1,2],[3,4]], B=[[5,6],[7,8]]

C[1][1] = 1×5 + 2×7 = 5+14 = 19

C[1][2] = 1×6 + 2×8 = 6+16 = 22

C[2][1] = 3×5 + 4×7 = 15+28 = 43
C[2][2] = 3×6 + 4×8 = 18+32 = 50
So C = [[19,22],[43,50]]

Transpose of a Matrix

Interchange rows and columns. If A is m×n, then A^T is n×m.
If A = [[1,2,3],[4,5,6]], then A^T = [[1,4],[2,5],[3,6]]

8.4 Inverse of a Matrix

For a 2×2 matrix A = [[a,b],[c,d]], the inverse A^(-1) = (1/det(A)) × [[d,-b],[-c,a]]
Condition: det(A) ≠ 0 (matrix must be non-singular)
Example: A = [[2,1],[5,3]]
det(A) = (2)(3)-(1)(5) = 6-5 = 1
A^(-1) = (1/1) × [[3,-1],[-5,2]] = [[3,-1],[-5,2]]

8.5 Properties of Matrices

Property	Formula
Commutative for Addition	A + B = B + A
Associative for Addition	(A+B)+C = A+(B+C)
NOT commutative for multiplication	AB ≠ BA (in general)
Associative for multiplication	(AB)C = A(BC)
Distributive	A(B+C) = AB + AC
Transpose of product	(AB)^T = B^T × A^T
Identity matrix	AI = IA = A
Inverse	AA^(-1) = A^(-1)A = I

Must Remember

• Matrix multiplication: columns of first = rows of second.
• AB is generally NOT equal to BA.
• Identity matrix I: AI = IA = A.
• Singular matrix: det=0, NO inverse. Non-singular: det≠0, HAS inverse.
• Symmetric: A = A^T. Skew-Symmetric: A^T = -A.
• (AB)^T = B^T × A^T (note the reversed order!).

Exam Questions

• Define matrix. Explain different types of matrices. (6 marks)
• Explain matrix multiplication with example. (4 marks)
• Find inverse of a 2×2 matrix. (4 marks)
• State properties of matrix multiplication. (4 marks)
• If A=[[1,2],[3,4]], find A^T and verify (A^T)^T = A. (4 marks)

QUICK REVISION SUMMARY — Subject 3: Basic Mathematics

Topic	Key Points
Statements	Declarative sentence: True or False. Simple vs Compound.
Connectives	~(NOT), ∧(AND), ∨(OR), →(IF-THEN), ↔(IFF).
AND	TRUE only when BOTH true.
OR	FALSE only when BOTH false.
IF-THEN	FALSE only when P=T and Q=F.
Tautology	ALWAYS TRUE. All T in truth table column.
Contradiction	ALWAYS FALSE. All F in truth table column.
Sets	Well-defined collection. Types: Empty, Finite, Infinite, Universal, Power Set.
Set Operations	Union(∪), Intersection(∩), Difference(-), Complement('), Symmetric Diff(⊕).
De Morgan's Laws	(A∪B)'=A'∩B' and (A∩B)'=A'∪B'.
Cartesian Product	A×B = set of all ordered pairs (a,b). |A×B| = |A|×|B|.
Determinants	2×2: ad-bc. 3×3: Cofactor expansion. Singular: det=0.
Cramer's Rule	x=Dx/D, y=Dy/D, z=Dz/D. Requires D≠0.
Matrix Types	Row, Column, Square, Identity, Diagonal, Symmetric, Triangular.
Matrix Multiplication	Columns of A must = Rows of B. (AB)^T = B^T × A^T.